

Overshoot Loss Proposal

Request for Senior Dev Review

Date: May 10, 2026 | Status: Discussion document, no changes implemented yet

Summary

After implementing the calibration filter and price cap (May 8), our loss pattern shifted. Overshoots now dominate. We need a strategy to address them without breaking what's working.

This document presents the data, four proposed solutions, and asks for the senior dev's input on which to pursue.

The Loss Pattern Has Shifted

Original system (pre-filter, all trades):

- 22 undershoots (65% of losses) — temperature kept rising after lock
- 12 overshoots (35%) — delta correction landed us one bracket too high

New system (calibrated cities, price < 30 cents, real-money trades only):

- 2 undershoots — \$90 in losses
- **10 overshoots — \$450 in losses**

The recently deployed fixes (rate-of-change check, cloud cover boost) target undershoots. They will help with \$90 of losses but do nothing for the dominant \$450 problem.

What An Overshoot Looks Like

For each real-money loss, here is the lock-time data. Note: some delta values shown are current; historical deltas may have differed at trade time.

Date	City	Lock max	Delta	Adjusted	Bet	Actual	Result
May 5	Hong Kong	23.2 C	-0.23	23.0 C	24 C	23 C	Over +1
May 5	Helsinki	13.2 C	-1.13	12.1 C	13 C	12 C	Over +1
May 6	Hong Kong	25.4 C	-0.23	25.2 C	25 C	24 C	Over +1
May 7	Madrid	18.1 C	-0.10	18.0 C	19 C	17 C	Over +2
May 8	Seoul	19.0 C	+0.33	19.3 C	21 C	20 C	Over +1
May 8	Tel Aviv	25.0 C	0.00	25.0 C	26 C	25 C	Over +1

Date	City	Lock max	Delta	Adjusted	Bet	Actual	Result
May 8	Amsterdam	17.0 C	+0.30	17.3 C	19 C	17 C	Over +2
May 9	Amsterdam	18.6 C	+0.30	18.9 C	19 C	18 C	Over +1
May 10	Wuhan	28.0 C	+0.80	28.8 C	29 C	28 C	Over +1
May 10	Chengdu	31.4 C	+0.72	32.1 C	32 C	31 C	Over +1

Two patterns visible

Pattern A — Boundary proximity (5 of 10): The adjusted temperature was within 0.3 C of a bracket boundary. Examples: Wuhan adj 28.8 (boundary at 28.5), Chengdu adj 32.1 (boundary at 31.5), Amsterdam adj 18.9 (boundary at 18.5).

Pattern B — Persistent positive bias (5 of 10): Cities with delta near zero or negative still overshoot, suggesting the METAR-to-WU relationship has structural noise we are not capturing. Tel Aviv (delta=0) overshoot. Hong Kong (delta=-0.23) overshoot twice.

Why Did the Previous Boundary Buffer Attempt Fail?

We previously implemented an upward boundary buffer that pushed our bracket selection to the next-higher bracket when within 0.5 C of the boundary. This caused immediate losses (Seoul, Tel Aviv on May 8) and was reverted.

That failure happened because the buffer pushed us *toward* the higher bracket. The current data suggests the opposite is needed: we should be more cautious near the *upper* boundary, not aggressive.

Four Proposed Solutions

Option 1 — Boundary Buffer (Skip Near Edge)

Skip trades when the adjusted temperature is within 0.3 C of a bracket boundary. Example: if adjusted = 28.8 C and the upper boundary is 28.5 C, we would skip rather than bet on bracket 29.

Pros:

- Directly addresses Pattern A (5 of 10 overshoots)
- Simple to implement
- No model retraining needed

Cons:

- Reduces trade count (estimated 30-40% of qualifying trades filtered)
- Doesn't help with Pattern B (intrinsic noise overshoots)
- Doesn't address overshoots that are clearly NOT near a boundary

Option 2 — Bayesian Shrinkage of Delta

For cities with small calibration samples ($n < 10$), shrink the estimated delta toward zero using a Bayesian prior. As n grows, trust the local delta more.

Formula: $\text{effective_delta} = (n / (n + K)) * \text{raw_delta}$, where K is a shrinkage parameter (suggested $K=5$).

Example with $K=5$:

- $n=3$ samples: use 37.5% of raw delta
- $n=5$ samples: use 50%
- $n=10$ samples: use 67%
- $n=50$ samples: use 91%

Pros:

- Statistically principled — addresses small-sample variance directly
- Self-correcting: cities prove themselves before getting full delta credit
- Recommended by senior dev in earlier review

Cons:

- Will increase undershoot rate for cities with strong positive deltas (Wuhan +1.0, Chongqing +1.0)
- Counterfactual analysis is mixed: only 4 of 10 overshoots prevented at 50% shrinkage

Option 3 — Delta Variance Tracking

Track the standard deviation of delta observations alongside the mean. Use $\text{mean} - K * \text{sigma}$ for prediction. Cities with high variance get smaller effective delta.

Implementation requires:

- Schema change to add `delta_variance` column
- Update resolver to maintain running variance
- Update prediction logic

Pros:

- Addresses Pattern B (cities with noisy deltas)
- More information-rich than shrinkage alone
- Aligns with senior dev's "delta-by-condition" idea (precursor)

Cons:

- More complex implementation (running variance)
- Needs minimum n before variance is meaningful (probably $n \geq 5$)
- Conservative bias may reduce overall trade count significantly

Option 4 — Higher Confidence Threshold

Raise `PHASE2_MIN_CONFIDENCE` from 0.81 to 0.90. Only trade when extremely confident (later in day, longer stability, or with cloud cover boost).

Pros:

- Trivial to implement (one config value)
- Reduces trade count and variance
- Naturally combines with cloud cover fix

Cons:

- Doesn't directly address overshoots — they happen at high confidence too
- May filter out high-EV cheap-bracket opportunities
- Crude tool

Combined Approach (My Recommendation)

The cleanest path forward might be combining Options 1 and 2:

1. **Boundary buffer with asymmetry:** Skip trades when adjusted temp is within 0.3 C of the *upper* bracket boundary (overshoot risk side). Don't apply on the lower boundary side.
2. **Bayesian shrinkage** with $K=5$ to address small-sample delta variance.

Reasoning:

- Boundary buffer attacks the most identifiable failure mode (Pattern A)
- Shrinkage addresses Pattern B more gradually
- Combined effect: skip ~30% of trades, but those skipped have the highest overshoot risk

Counterfactual Backtest Limitations

I want to be honest about what I can and cannot backtest:

- **What I have:** Lock-time temperature, current delta values, actual resolution outcomes, bet bracket
- **What I do not have:** Historical delta values at the time of trade (deltas have evolved), raw temperature reading history, day-by-day market price snapshots

This means counterfactual estimates are directionally useful but not precise. A proposed change that "saves 4 of 10 overshoots" in backtest might save 2 or 6 in production.

The most reliable test is forward results. We should pick one approach, run it for 14 days, and compare to the current 35.7% / +\$1,476 baseline.

Specific Questions for Senior Dev

1. **Is asymmetric boundary buffer (Option 1) defensible?** It treats overshoots as worse than undershoots, which is true under our payout structure (cheap brackets) but feels ad-hoc.
2. **Bayesian shrinkage (Option 2) — what K value?** I suggested $K=5$. Too aggressive? Too conservative? What's the typical convention for small-sample bias correction in similar trading systems?

3. **Should we track delta variance (Option 3)?** It's the most informative approach but requires schema changes and minimum samples. Worth the complexity?
4. **Are these fixes mutually exclusive or stackable?** I assume stackable but want validation.
5. **What's the right success metric?** I'm proposing 14 days of forward data vs. May 2-10 baseline. Is that enough? Would you suggest a different evaluation method?
6. **Any hidden risks I'm missing?** Each fix reduces trade count, which means longer time to statistical significance. Is there a way to address overshoots that doesn't reduce sample size?

Additional Context: What's Already Live

Three fixes were deployed today (May 10):

1. **Bracket matching fallback** — Jakarta/KL/Shanghai now find nearest available bracket within 1 C / 2 F when exact match fails (purely additive, no risk)
2. **Rate-of-change check** — Lock requires flat or declining trend over last 6+ readings (targets undershoots only)
3. **Cloud cover boost** — BKN/OVC sky conditions add +0.05/+0.08 to lock confidence (targets undershoots, marginal)

None of these address the overshoot problem this document is about.

Bankroll Status

Metric	Value
Starting bankroll (May 2)	\$1,000
Current bankroll (May 10)	\$2,415
Real-money trades placed	17
Win rate	29.4%
Total P&L	+\$1,341
ROI on deployed	175.3%

The system is profitable. We are seeking incremental improvement, not rescue.